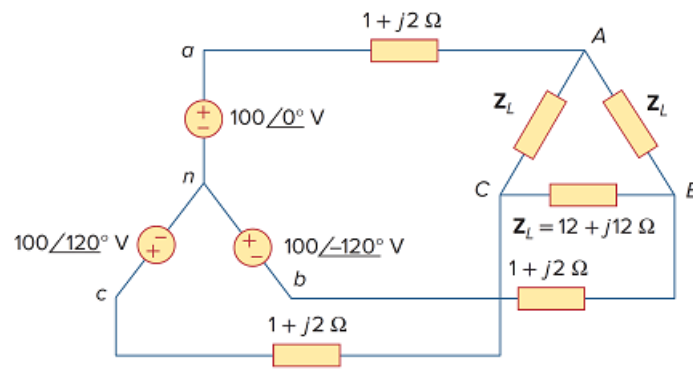


Problem

Obtain the line currents in the three-phase circuit of Fig. 12.47 on the next page.

Figure 12.47



Step-by-step solution

Step 1 of 7

Refer to Figure 12.47 in the textbook.

From the three-phase circuit, it is clear that source is Wye connected and load is delta connected. First convert the Δ connected load impedance to Y connected load impedance.

The load impedance in the equivalent star connection is,

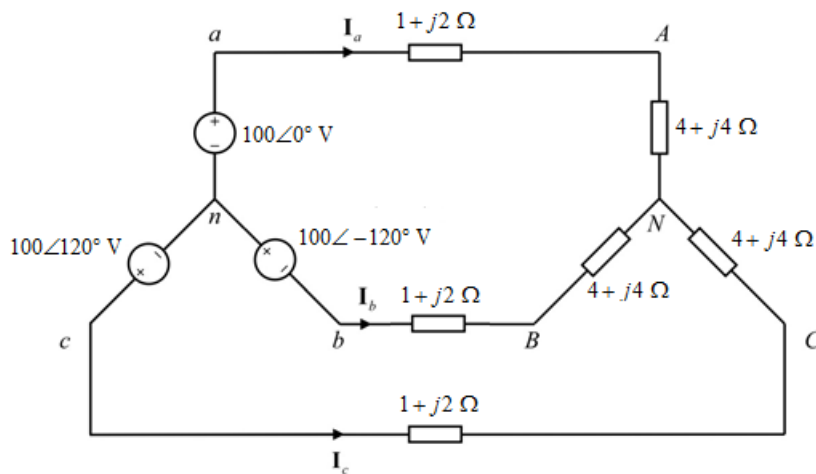
$$\begin{aligned} Z_{YL} &= \frac{Z_L \cdot Z_L}{Z_L + Z_L + Z_L} \\ &= \frac{Z_L^2}{3Z_L} \\ &= \frac{Z_L}{3} \end{aligned}$$

Step 2 of 7

Determine the equivalent load impedance in star connected load.

$$\begin{aligned} Z_{YL} &= \frac{Z_L}{3} \\ &= \frac{12 + j12}{3} \\ &= 4 + j4 \Omega \end{aligned}$$

Draw the modified three phase Wye-wye circuit diagram.



Step 3 of 7

The phase voltages of balanced Y-Y four wire system are,

$$\begin{aligned} V_{an} &= 100\angle 0^\circ \text{ V} \\ V_{bn} &= 100\angle -120^\circ \text{ V} \\ V_{cn} &= 100\angle 120^\circ \text{ V} \end{aligned}$$

The load impedance per phase, Z_{YL} is $4 + j4 \Omega$

The line impedance per phase, Z_l is $1 + j2 \Omega$

Step 4 of 7

Calculate the value of total load impedance per phase, Z_Y .

$$Z_Y = Z_{YL} + Z_l$$

Substitute $4 + j4 \Omega$ for Z_{YL} and $1 + j2 \Omega$ for Z_l .

$$\begin{aligned} Z_Y &= 4 + j4 + 1 + j2 \\ &= 5 + j6 \Omega \end{aligned}$$

Step 5 of 7

Calculate the line current of balanced Y-Y four wire system, $\mathbf{I}_a$.

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}$$

Substitute $5 + j6 \, \Omega$ for $\mathbf{Z}_Y$ and $100\angle 0^\circ \text{ V}$ for $\mathbf{V}_{an}$.

$$\begin{aligned}\mathbf{I}_a &= \frac{100\angle 0^\circ}{5 + j6 \, \Omega} \\ &= \frac{100\angle 0^\circ}{7.81\angle 50.2^\circ} \\ &= 12.8\angle -50.2^\circ \text{ A}\end{aligned}$$

Step 6 of 7

Calculate the line current of balanced Y-Y four wire system, $\mathbf{I}_b$.

$$\mathbf{I}_b = \frac{\mathbf{V}_{bn}}{\mathbf{Z}_Y}$$

Substitute $5 + j6 \, \Omega$ for $\mathbf{Z}_Y$ and $100\angle -120^\circ \text{ V}$ for $\mathbf{V}_{bn}$.

$$\begin{aligned}\mathbf{I}_b &= \frac{100\angle -120^\circ}{5 + j6 \, \Omega} \\ &= \frac{100\angle -120^\circ}{7.81\angle 50.2^\circ} \\ &= 12.8\angle -170.2^\circ \text{ A}\end{aligned}$$

Step 7 of 7

Calculate the line current of balanced Y-Y four wire system, $\mathbf{I}_c$.

$$\mathbf{I}_c = \frac{\mathbf{V}_{cn}}{\mathbf{Z}_Y}$$

Substitute $5 + j6 \, \Omega$ for $\mathbf{Z}_Y$ and $100\angle 120^\circ \text{ V}$ for $\mathbf{V}_{cn}$.

$$\begin{aligned}\mathbf{I}_c &= \frac{100\angle 120^\circ \text{ V}}{5 + j6 \, \Omega} \\ &= \frac{100\angle 120^\circ \text{ V}}{7.81\angle 50.2^\circ} \\ &= 12.8\angle 69.8^\circ \text{ A}\end{aligned}$$

Therefore, the line currents in a balanced Y-Y four wire system are,

$\begin{aligned}\mathbf{I}_a &= 12.8\angle -50.2^\circ \text{ A} \\ \mathbf{I}_b &= 12.8\angle -170.2^\circ \text{ A} \\ \mathbf{I}_c &= 12.8\angle 69.8^\circ \text{ A}\end{aligned}$
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